

Experimental Characterization of Synchronization Protocols for Instrument Wireless Interface

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Abstract—Nowadays, careful measurement applications are handed over to wireless sensor networks (WSNs). In most of these tasks, sensor nodes work together to recognize the data fusion process. Synchronization is a critical element in this scenario. Nodes have to be regulated to a common clock and regulated among them. Many synchronization algorithms can be found in literature, and some of them have been intentionally established for low-cost structural designing where efficient memory management and reduced computational burden are important constraints. The performance of these algorithms is usually evaluated either in simulation environments or by adopting ad hoc radio systems. Attention is rarely paid to the influence of factors, such as the following: clock stability, the dependence of the measurement application on the synchronization algorithm and vice versa, the effect of the limited bandwidth available on real radio systems, and the change of performance when common commercial wireless communication busses such as Wi-Fi, Bluetooth (BT), and ZigBee are adopted. This paper makes the first step toward this direction, proposing the experimental characterization of a small-scale WSN implementing some of these protocols versus common wireless busses such as BT, Wi-Fi, and ZigBee.

Index Terms—Clock synchronization, measurement, time synchronization, triggering, wireless sensor network (WSN).

I. INTRODUCTION

WIRELESS sensor networks (WSNs) today are understood as networks of tiny low-power devices capable of performing onboard sensing, computing, and communication tasks in areas without any infrastructure [1], [2]. The smart sensors connected to the WSNs respond to atypical requirements (e.g., low cost and reliability) and suffer unavoidable constraints, such as limited energy resources, etc. Nevertheless, WSNs are finding places in a wide range of applications. In particular, WSNs can be divided into two categories. One includes large-scale networks, which are composed of hundreds or thousands of nodes that operate in large geographical areas. These networks perform very well in applications, such as communication, military, and environment monitoring. The other represents the small-scale network category, composed of tens of sensors. Measurement applications, such as remote mea-

surement laboratories and monitoring of industrial processes, to name a few, belong to this second category.

In most of these applications, the efficiency of a WSN vastly depends on the time synchronization among its nodes [3]. The coordination among these nodes for power-saving sleep/wakeup modes, synchronized measurements, data fusion, localization of sensor nodes, and object tracking requires that all of the nodes are running on a common time frame. These requirements cannot be met by conventional network synchronization protocols by the Internet, due to the limited energy resource of nodes and the low-complexity requirements of WSNs [3].

More suitably, the time synchronization problem in WSNs can be broken up into two steps, namely: 1) the adjustment of the clock time phase offset (clock offset) among the nodes to a common absolute time; and 2) the correction of the clock frequency offset (clock skew) with respect to a given reference frequency. Since imperfections in the quartz crystal and changes in the environmental conditions cause different clocks to run at slightly different frequencies, the effect of the clock skew is the main reason why clock offsets keep drifting apart. Therefore, the continuous adjustment of the clock offset by itself allows the time synchronization among the nodes to be achieved. Adjusting also the clock skew would guarantee a long-term reliability of synchronization and therefore reduce network-wide energy consumption in synchronization protocols. Of course, clock skew adjustment cannot simultaneously be extremely accurate and also have low energy consumption. For these and other reasons, developing long-term and network-wide time synchronization protocols that are energy efficient represents one of the key strategies for the successful deployment of long-life WSNs.

The time synchronization of network nodes is also a problem arising in particular measurement applications when the instruments are equipped with wireless modules, as well as when the instruments are featured with suitable wired or wireless interfaces. In these cases, the wireless connection prevents the possibility of hardware triggering that is present in many wired busses [4].

Literature today is dealing with this synchronism in both serial search strategy problems [5], [6] and in efficient algorithms aiming at WSN synchronization [7]–[12].

As for the first issue, innovative papers have demonstrated that an optimized acquisition strategy may improve all the synchronization problems reducing the overall synchronization time.

As far as the second issue is concerned, some algorithms, such as the IEEE 1588 PTP [13], are characterized by complex

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protocols and may grant a very interesting performance achieving tens of picoseconds. Others give a chance for a suitable implementation in cost-effective designs where low memory requirement and acceptable computational burden are insurmountable constraints. Algorithms, such as the Timing Synch Protocol for Sensor Networks (TPSN) [7], Reference Broadcast Synchronization (RBS) [8], Flooding Time Synch Protocol (FTSP) [9], Broadcast Synchronization Over Bluetooth (BT) [10], Adaptive-Rate Time Synchronization Protocol [11], Time Synchronization with Extended Clock Model [12], and many others have been intentionally designed for low-power WSNs and seem suited for the purpose.

Unfortunately, most of these algorithms were tested either in simulation environments or by adopting ad hoc radio systems. Attention has rarely been paid to the influence of factors, such as the following:

- (i) the change of performance if low cost and general purpose hardware is linked together with the influence of the clock stability;
- (ii) the influence of the measurement application on the synchronization algorithm and vice versa;
- (iii) the effect of the limited bandwidth available on real radio system and the change of performance when common commercial wireless communication busses such as Wi-Fi, Bluetooth (BT), ZigBee are adopted;

As far as the first point is concerned, many synchronization algorithms foresee time estimations based on linear regressions, averages, polynomial fittings, etc. In many cases, the time required to complete these operations is not considered, since the computational burden is supposed to be very small. The authors think that when these algorithms are implemented on real sensor nodes, the required time may be high enough depending on the adopted hardware and the clock rate. This time that stands for an accumulation error strictly influences the synchronization performance particularly if nodes involved in the synchronization procedure have got clocks that exhibit sensible drifts. In this case, the obtained synchronization performance is worsened by an intrinsic offset related to the computational burden even if the algorithm tries to compensate the drift.

As for the second point, many proposals consider the generation of time stamps at the Medium Access Channel (MAC) level. The common implementations of these algorithms are, instead, made at the application level. In this case, the generated time stamps may be affected by latencies due to the contemporary measurement application that could be engaged in the execution of critical measurement tasks.

The third point considers that every real radio channel shows a limited transmission bandwidth together with the intrinsic transmission latency due to the time spent to drive the data from the application to the transmission layer which reduces the performance of synchronization algorithms. This is truer if other commercial wireless busses, such as Wi-Fi, BT, ZigBee, and others are considered. In this case, the policies of channel access and negotiations might deeply influence the performance of synchronization algorithms.

For these reasons, a lack of both metrics to assess and compare the performance of these algorithms and experimental evaluations of the performance bounds is evident.

The authors have been working with this research field for a few years, proposing solutions for both power-aware devices operating in WSNs [14] and wired to wireless interface systems for instruments [4], [15].

From their previous experience, they have provided the first functioning analysis on two of the most common synchronization protocols (TPSN and FTSP) when involved in real WSN, where the cost effectiveness of the hardware and the aforementioned constraints have to be considered [16].

In this paper, the authors expand their previous studies by providing an experimental characterization on small-scale WSNs of the considered synchronization protocols. The first aim of this paper is featuring a wireless interface of instrumentation and/or sensor with accurate triggering resources which could be important in distributed measurement applications. Then, as a further result, it may also provide useful indications in selecting the synchronization algorithm in case of a low number of sensors (tens) and a small geographical scale of the network. In particular, this paper will be focused on the aforementioned first and third points to analyze the change of performance if low-cost and general-purpose hardware is provided and the effect of the limited bandwidth available on a real radio system, such as Wi-Fi, BT, and ZigBee.

Further study will concern the characterization of more complex protocols, such as the IEEE 1588 PTP, the analysis of the second point effects, and the influence of polluted environments, such as those of industrial applications.

II. THEORETICAL BACKGROUND

Different types of synchronization protocols are present in literature. They operate according to several strategies, namely: 1) master-to-slave or peer-to-peer architectures; 2) clock drift correction or clock correction independence; 3) inside or outside network synchronization; 4) sender-to-receiver or receiver-to-receiver synchronization; 5) one- or two-way messaging; and 6) multihop or single-hop networks. Nowadays, a commonly used approach is based on the sender-to-receiver or receiver-to-receiver synchronization.

The sender-to-receiver synchronization approach usually operates as follows. The sender node sends a time stamp message to the receiver one; the receiver can modify its clock or not (in hardware or software mode) by using the arrived time stamp; the receiver sends its time stamp to the sender that calculates and compensates for the clock phase delay of the receiver. This procedure is repeated for every receiver present in the network. The time that the sender time stamp message takes to reach the receiver one is calculated by measuring the total time of the synchronization procedure and dividing that by two. Obviously, this time is not deterministic, since it is dependent on network delays and the node tasks. Some sender-to-receiver synchronization protocols estimate this time by calculating the mean time over a number of consecutive transmissions, while for others, the sender transmits a time stamp that is its actual absolute time. The former approach makes the computational

burden of the nodes more serious, while the latter is affected by a nondeterministic error related to the time that the message spends to reach the receiver node. This quantity is usually called the time-critical path.

The receiver-to-receiver synchronization makes use of broadcast messaging. Two or more receivers get the same message from a sender and interact by exchanging the values of the local clock time at which they received the sender message. Then, they calculate their offsets with respect to the reception times. An advantage of this method is the cutting down of the critical time path, since the sender disappears from the synchronization procedure.

In the following, the two most important synchronization algorithms, the TPSN and the FTSP, that belong to the approaches mentioned earlier, are described in detail. As far as the RBS protocol is concerned, it is worth noting that it was designed to overcome some limits of the FTSP in terms of computational burden, when numerous nodes (more than 50) are present in the network. Since this paper would not have involved more than ten nodes, the people analyzing the algorithms thought that it was ineffective to test a protocol that was not expected to exhibit a difference in performance in relation to the FTSP.

A. TPSN

TPSN [7], proposed by Ganeriwal *et al.*, is based on a hierarchical WSN design, in which each node communicates only with nodes that belong to lower levels and performs the time synchronization based on a sender-to-receiver approach. In the first phase, called level discovery, the root node looks for nodes that belong to lower levels. Once created, the hierarchical composition of the synchronization phase starts from the root node and reaches each node in all of the levels below. Nodes in the middle act as gateways for the lower nodes.

Then, pairwise synchronization is performed along the edges of the hierarchical structure established in the earlier phase according to the classical approach of the sender-to-receiver synchronization. Fig. 1 shows this message-exchange between nodes A and B. In Fig. 1, T1 and T4 represent the time values measured by node A, whereas T2 and T3 represent the time values measured by node B. At time T1, A sends a synchronization packet to B. The synchronization packet contains the level number of A and the value of T1. Node B receives this packet at T2, where T2 is equal to $T1 + \Delta + d$. Here, Δ represents the clock phase drift between the two nodes, whereas d represents the propagation delay. At time T3, B sends back an acknowledgment packet to A. The acknowledgment packet contains the level number of B and the value of T1, T2, and T3. Finally, node A receives this other packet at T4. Assuming that the clock phase drift and the propagation delay do not change in this small span of time, A can calculate the clock drift and propagation delay as

$$\Delta = \frac{[(T_2 - T_1) - (T_4 - T_3)]}{2}d = \frac{[(T_2 - T_1) + (T_4 - T_3)]}{2} \quad (1)$$

Knowing the drift, node A can correct its clock accordingly, so that it synchronizes with node B. This is a sender-initiated

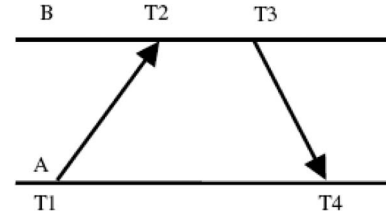


Fig. 1. Example of a two-way messaging between nodes A and B. T2 and T3 are measured in node B, while T1 and T4 are measured in node A.

approach, where the sender synchronizes its clock to that of the receiver.

Characterized by low computational burden and simple implementation, this protocol needs two-way message signaling that increases the data sent out. The computational burden rises corresponding to the number of nodes, since each node must synchronize itself with a reference node.

As far as the performance of the TPSN algorithm is concerned, the proposal writers in the emulation environment measured a residual mean error of about $17 \mu\text{s}$, whereas in their tests made on ad hoc hardware, such as the Xbow Mica Mote device, and over a long observation period, Yoon *et al.* permit an average error of about $4.8 \mu\text{s}$ [17].

B. FTSP

The goal of the FTSP [11], proposed by Maróti *et al.*, is to achieve network-wide synchronization between the local clocks of the participating nodes. Since typical WSN operates in areas larger than the broadcast range of a single node; as a result, the FTSP provides multihop synchronization. The root of the network—a single dynamically (re)elected node—maintains the global time, and all of the other nodes synchronize their clocks to the one of the root. The nodes form an ad hoc structure that transfers the global time from the root to all of the other nodes, as opposed to the proposed fixed spanning-tree-based approach. This saves the initial phase of establishing the tree, and it is stronger against node and link failures and dynamic topology changes.

Assuming that each node has a local clock exhibiting the typical timing errors of crystals and can communicate over an unreliable but error-corrected wireless link to its neighbors, the FTSP synchronizes the time of a sender to multiple possible receivers utilizing a single radio broadcast message time stamped at both the side of the sender and the receiver. MAC layer time stamping can eliminate many of the errors, as observed in [9]. However, accurate time-synchronization at discrete points in time is only a partial solution. Compensation for the clock drift of the nodes is inevitable to achieve high precision in between synchronization points and to keep the communication overhead low. Linear regression is used in FTSP to compensate for clock drift.

A message dispatched from the sender has got the sender time stamp. Each receiver compares the received time stamp with that obtained from its local clock at the receiving time. Since senders and receivers show a drift in their local clocks, a continuous synchronization process, with time updating at lower than 1 s, is required to have errors in some microsecond

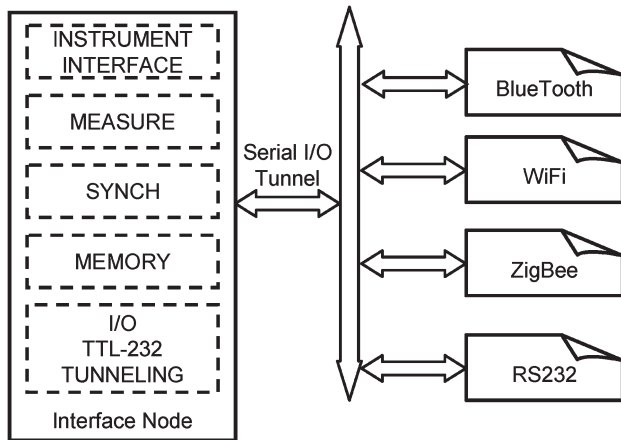


Fig. 2. Architecture of the wired to wireless interface boards.

range. Since typical drifts are linear, a regressive pattern may predict and offset errors thus reducing the updating rate. This protocol has the advantages of having a low transmission loss and being robust at the death of some nodes, but it has the disadvantage of collecting more time stamps putting into effect a linear regression. As far as the performance is concerned, the implementation on ad hoc hardware, such as the Xbow Mica 2 device, grants an average error of about $2 \mu\text{s}$.

III. MEASUREMENT SETUP

A suitable measurement network representing a small-scale WSN and two measurement setups has been completed to evaluate the performance of the aforementioned algorithms.

A. Wireless System Network

The wireless interface network is made up of a number of wired to wireless devices that are able to grant Wi-Fi or BT connectivity to measurement systems by not embedding a suitable antenna. For each one of the considered wireless technologies, a wireless star bus can be carried out by a master module hosted in a PC controller that grants BT and Wi-Fi connections, while slaves are wired to wireless interfaces that fit the RS232, IEEE488, or USB ports of the measurement systems. Fig. 2 shows the architecture of each wired to wireless interface board, while Fig. 3 shows a diagram of the completed measurement bus in the situation of a BT connection.

All of the wired to wireless interface devices are based on the low-power Microchip PIC18LF4620 microcontroller. It allows a maximum frequency of 40 MHz, 32 kB of program memory, 1536 B of RAM memory, one USART serial port, and one I2C bus. By using the onboard USART TTL232 port, a serial channel is recognized, and the connection with the wireless BT, Wi-Fi, and radio devices is allowed. All of the wired to wireless devices are also equipped with a commercial clock with a nominal frequency of 8 MHz and an absolute accuracy of 60 ppm in normal environmental conditions.

As far as the BT connection is concerned, a Parani ESD 110 device has been adopted. It allows communicating wirelessly throughout the range of $100 \sim 1000$ m with a BT version 1.2

protocol. The connection with the microcontroller is granted by a serial TTL port operating at 115 200 b/s.

The Wi-Fi connection is carried out by the EZL-80C universal RS232 (TTL) converter for a wireless data transfer via Wi-Fi 802.11b. The module is connected by a Wi-Fi card (Ambicom wireless LAN flash card) containing the radio component. A built-in EZL-80 converter supports conversion from a TTL serial line to a standard Wi-Fi line (wireless LAN) IEEE802.11b, which maintains wireless Ethernet in the frequency band of 2.4 to 2.4835 GHz by a direct-sequence spread spectrum modulation technique.

As far as software development is concerned, the ANSI C language and the Mikro C environment provided by MikroElektronika have been adopted. Since all of the considered algorithms require a timer that is able to generate time stamps and a clock reference, the built-in timer of the microcontroller and the internal clock references have been considered. To debug the algorithms performance, suitable I/O pins from the microcontrollers have been selected to generate pulse or square wave signals. The implemented algorithms for the TPSN and FTSP protocols are described in [18].

B. Setup for the Experimental Characterization of the Interface Node Clocks

Since the drift and the skew of the node clock can influence the whole synchronization procedure along with the time of resynchronization, all of the node clocks have been accurately characterized by means of the measurement station in Fig. 4.

The PC bus controller is linked, by the way of the IEEE 488 interface, to an Agilent 53132A digital counter and to a LeCroy WaveMaster 8620A Digital Scope. The digital counter and scope, equipped with a special jitter measurement capability, allow the possibility to assess the performance of each board by evaluating the clock performance in terms of clock frequency, clock skew, and histogram. The clock frequency and skew have been estimated by over 1000 tests, while up to one million tests were considered for the histogram evaluation.

C. Setup for the Experimental Characterization of the Synchronization Protocols

A suitable measurement station has been designed to characterize the synchronization algorithms (Fig. 5). It is made up of the PC controller, a Tektronix TLA5203 Logic Analyzer, and a number of digital multimeters connected to the respective wireless interface nodes. The logic analyzer, by the way of suitable debug pins, is able to estimate the time the synchronization event starts, the duration of the synchronization, and the clock phase delays, understood as the misalignment between the wireless interface nodes, before and after the synchronization procedure.

IV. EXPERIMENTAL RESULTS

Results of the experimental tests carried out to evaluate the performance of the TPSN and FTSP synchronization protocols are described in the following sections.

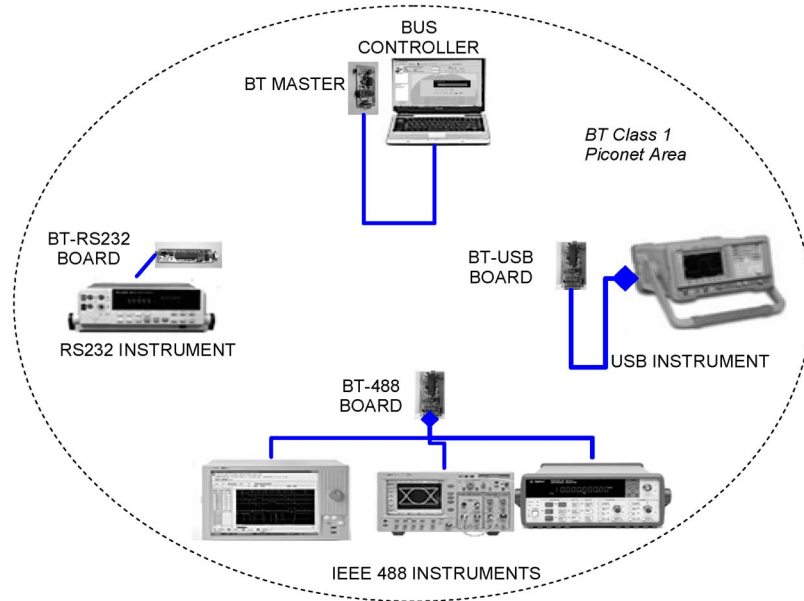


Fig. 3. Realized measurement bus. The BT radio is involved.

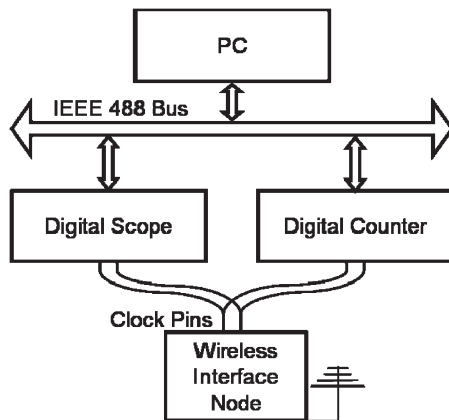


Fig. 4. Block diagram of the automatic measurement station for the characterization of the interface nodes.

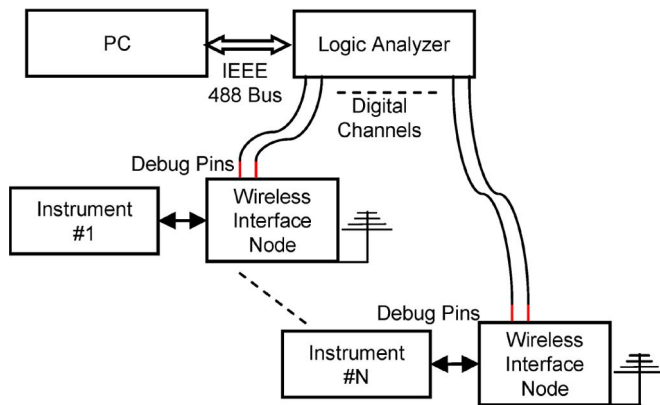


Fig. 5. Block diagram of the automatic measurement station for the characterization of synchronization procedure.

A. Organization of the Tests

Experiments have been carried out in two steps. At first, the clock of each node was properly distinguished, and the

aforementioned values were estimated. Successively, the performance of the synchronization algorithms had been assessed. Two sets of experiments have been carried out. In the first one, called Setup A, the network was made up of only two nodes (one sender and one receiver device) equipped with the wired and wireless interfaces considered earlier. In the second set of experiments, called Setup B, one sender and up to four receivers were involved. This allowed evaluating the receiver-to-receiver synchronization performance. Both Wi-Fi and BT wireless busses have been considered. As for the BT wireless bus, several transmission modes have been experimented with while considering the following: 1) synchronous and asynchronous data transfers; and 2) DM and DH packets. As far as the Wi-Fi is concerned, TCP-IP and UDP transmission modes have been considered. Results of all of the considered tests have also been compared with those provided by a RS232 wired connection operating at the same rate as the wireless busses that were considered. This allows evaluating the influence of the latencies of the wireless devices on the synchronization procedures.

To highlight the effects of the loss of synchronization, a square wave signal with a span of 2 s has been generated by each node. This signal is driven by an interrupt procedure based on some timer registers that are directly driven by the clock and therefore affected by the synchronization procedure performance. These signals are generated by suitable digital debug pins implemented on the interface boards and are acquired by the logic analyzer as shown in Fig. 5.

Suitable figures of merit have been introduced to assess the performance of the considered synchronization algorithms. They consider the following:

- the clock phase delay (CPD) that is expressed as the mean time interval between the rising edges of the square wave transitions of the sender and the receiver node clocks after the synchronization procedure (CPD_{MEAN});
- the relative experimental deviation of the CPD figure of merit (σ_{CPD});

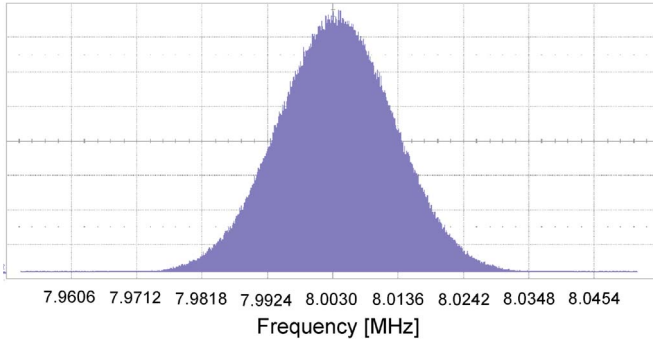


Fig. 6. Experimental histogram of the sender clock frequency. A mean value equivalent to 8.00062 MHz and a standard deviation of 9.43 kHz had taken place. More than one million measurements were involved.

TABLE I
SOME EXPERIMENTAL RESULTS RELATING TO THE CHARACTERIZATIONS OF THE INTERFACE NODE CLOCKS. A GATE TIME OF 60 s TOOK PLACE

Board under test	μ_F	σ_F	Δ_{CLOCK}
	[Hz]	[Hz]	[ps]
Sender Node	7999791.07	0.17	n/a
Receiver Node #1	7999771.42	0.32	-0.31
Receiver Node #2	8000297.14	0.20	7.91
Receiver Node #3	8000267.81	0.37	7.45
Receiver Node #4	8000230.69	0.24	6.87

- c) the histogram of the relative distribution of the CPD over 100 experiments (H_{CPD});

B. Experimental Characterization of the Interface Node Clocks

The time required by one synchronization is typically bounded in tens of milliseconds, while the time between two consecutive resynchronizations may be up to a dozen (or dozens) of seconds. For this reason, the performance of the wireless interface node clocks has been properly characterized in these two times. First, the histogram of the clock frequency of the considered nodes has been determined on the basis of the results of more than a million measurements executed by a digital scope (Fig. 6). All of the tested clocks have shown a normal distribution.

Successively, by a digital counter, the mean value of the clock frequency (μ_F) and its experimental standard deviation (σ_F), have been measured with gate times of 1 ms, 10 s, and 60 s. Very similar results have been obtained for all the considered tests. Some of the obtained results are reported in Table I, where μ_F and σ_F , respectively represent the mean value and the experimental standard deviation of the clock frequency, and Δ_{CLOCK} is the mean value of the difference between the clock period of a considered node with respect to a sender. In the whole test, a gate time equivalent to 60 s is regarded. It can be noted that even if all of the clocks respect their nominal accuracies, a maximum Δ_{CLOCK} of 7.91 ps is experienced between the sender and the receiver#2 node. This clock delay leads to a clock drift equal to 632 μs if an observation time of 10 s is considered. This drift represents an in-

TABLE II
RESULTS OF THE EXPERIMENTAL CHARACTERIZATIONS OF THE TPSN SYNCHRONIZATION ALGORITHM. THE WIRED RS232 CONNECTION IS CONSIDERED

Baud Rate	synchronization Interval	CPD	σ_{CPD}
[kbps]	[s]	[ms]	[ms]
9600	5	0.633	0.019
	10	0.631	0.020
	30	0.632	0.020
19200	5	0.340	0.017
	10	0.339	0.015
	30	0.339	0.015
57600	5	0.150	0.013
	10	0.148	0.012
	30	0.148	0.012
115200	5	0.094	0.014
	10	0.097	0.015
	30	0.097	0.015

trinsic worsening of the synchronization, since different nodes have to execute the same computational load to perform the synchronization task, and they will take up different amounts of time.

Finally, the clock drift also influences the range of the synchronization intervals. Once a suitable threshold in the clock delay has been chosen, the more the drift, the shorter the time is between consecutive synchronizations.

The measured values of clock accuracy and repeatability have been used to correct the performance of the analyzed synchronization protocols.

C. Experimental Characterization of the Synchronization Protocols

Results pertaining to the performance of the TPSN and FTSP synchronization protocols are reported in the following.

At first, Setup A has been considered, and the RS232 wired connection has been involved. The sender and the receiver#1 devices have been involved. The results are reported in Tables II and III for the TPSN and the FTSP, respectively. In particular, the tests are arranged considering 100 consecutive synchronization procedures that were repeated every 5, 10, and 30 s. In addition, the RS232 baud rate has been changed from 9600 to 115 200 b/s.

Looking at the obtained results, some considerations can be made, namely:

- In any case the TPSN algorithm offers better performance values than the FTSP, when considering the all of the figures of merit. In depth the acceptable value of the σ_{CPD} figure confirms the credibility of the obtained results.

TABLE III
RESULTS OF THE EXPERIMENTAL CHARACTERIZATIONS OF THE FTSP
SYNCHRONIZATION ALGORITHM. THE WIRED RS232
CONNECTION IS CONSIDERED

Baud Rate [kbps]	synchronization Interval [s]	CPD [ms]	σ_{CPD} [ms]
9600	5	7.374	0.48
	10	7.375	0.48
	30	7.371	0.49
19200	5	3.875	0.26
	10	3.871	0.27
	30	3.879	0.29
57600	5	1.545	0.14
	10	1.543	0.15
	30	1.544	0.15
115200	5	0.967	0.12
	10	0.968	0.13
	30	0.968	0.13

- b) All the considered synchronization protocols show the worst performance when compared with those achieved in simulation and emulation environments [7]–[9];
- c) The considered synchronization protocols are strongly affected by the considered baud rate. In particular, the higher the baud rate the lower the CPD value;
- d) both of the synchronization protocols are not affected by the considered synchronization interval in all of the selected valuable reports.

Some factors influence the obtained results. They are concerned with the synchronization protocols implementation, the Δ_{clock} of the considered nodes, and the latency and bandwidth of the transmission medium. In particular, the time required to make a synchronization between the two nodes can be considered as the sum of the processing and the transmissions times. The former depends on the synchronization algorithm and its implementation, the latter depends on the number of transmission, the bandwidth, and the latency of the transmission medium.

As far as the TPSN algorithm is concerned, it requires the exchange of four messages per synchronization. These four time stamps are then managed according to (1). The implementation made by the authors uses a 32-b word for each time stamp transmission [18]. This means that the time spent to perform the TPSN synchronization algorithm can be considered as the sum of three times, namely: 2) the time required to transmit four words, each one 32 b long; 2) the time spent to calculate equations described in Section II-A; and 3) the time spent to execute microcontroller instructions (storage, computation, etc.).

As for the FTSP synchronization algorithm, it requires a large number of sender–receiver transmissions. In the implementation made by the authors, a total of eight sender–receiver transmissions is considered [18]. As done for the TPSN,

TABLE IV
RESULTS OF THE EXPERIMENTAL CHARACTERIZATIONS OF THE TPSN
AND FTSP SYNCHRONIZATION ALGORITHMS. THE WIRELESS BT
CONNECTION IS CONSIDERED

Baud Rate	CPD [ms]	σ_{CPD} [ms]
TPSN		
9600	10.3	3.1
19200	8.6	2.0
57600	7.3	3.4
115200	6.5	2.8
FTSP		
9600	42.0	2.8
19200	38.0	2.3
57600	34.3	2.8
115200	33.7	2.5

each time stamp transmission considers 32-b-long words. Each sender–receiver transmission is executed every 2 s. This means that a total time greater than 16 s is required to perform a synchronization procedure. Then, the time spent to perform the implemented version of the FTSP synchronization algorithm can be considered as the sum of four times, namely: 1) the time required to transmit 16 words, each one 32 b long; 2) the time spent to calculate a linear regression on the stored data; 3) the time spent to execute microcontroller instructions (storage, computation, etc.); and 4) the 16 s of total waiting.

It is then evident that the FTSP algorithm requires a total time that is greater than that of the TPSN. This time is due to the high waiting time among messages exchange, to the higher number of transmissions, and to the higher computational time of the FTSP.

Looking at comments reported in a), since the two considered nodes exhibit a clock drift, it is clear that the more time to perform the synchronization procedure, the more the accumulated clock error between nodes. This implies the wrong performance of the FTSP with respect to the TPSN. As for the results commented in b), since the aforementioned clock drift between the considered nodes and the transmission bandwidth and latency, the reached performance is very different from that expected in literature. As far as the c) comment is concerned, the higher the transmission rate, the lower the transmission time, the lower the overall synchronization time, and the lower the accumulated clock error in the nodes. As for d), the obtained results highlight as the performance of both synchronization algorithms are not affected by the initial value of the clock drift that is strictly related to the resynchronization time.

Further tests have been executed considering the BT wireless channel in which the sender is a BT slave operating in DH5 mode, and a connection/disconnection procedure per synchronization is executed. Results are reported in Table IV.

TABLE V
ANALYSIS OF THE INFLUENCE OF THE BT TRANSMISSION MODE ON THE TPSN ALGORITHM

<i>TPSN</i>	CPD_{MEAN} [ms]	σ_{CPD} [ms]	R_{CPD} [ms]
DH1	7.3	1.3	6.6
DH3	6.94	0.97	1.97

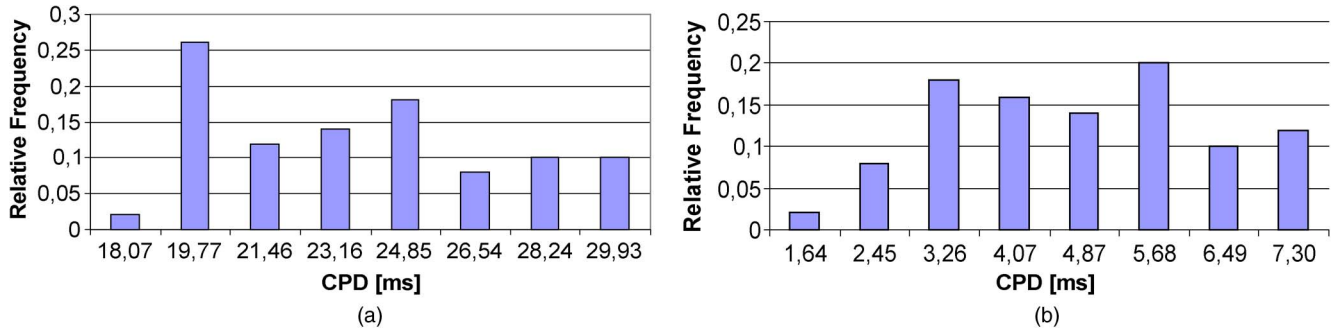


Fig. 7. Performance evaluation between the (a) FTSP and the (b) TPSN synchronization algorithms, when the Wi-Fi (TCP-IP) wireless connection is considered.

Some considerations can be addressed, namely:

- Also in this case the TPSN synchronization protocol shows a better performance in respect to the FTSP one even though the difference in the CPD figure of merit is lower than the wired RS232 case.
- For both the synchronization protocols the σ_{CPD} figure of merit increases in respect to the wired case. This highlights the increment of variability related to the use of BT channel.
- For both the synchronization protocols the higher the baud rate the lower the absolute delay, even though this effect is less prevailing than the wired RS232 case.

As for the f) and g) comments, they confirm what is stated for the wired RS232 case and demonstrate that the presence of real communication channels, real algorithm implementations, and cost-effective hardware causes worsening with respect to what is stated in literature. As far as the h) comment is concerned, the effect can be explained by looking at the total time required to transmit a BT packet. When two nodes are linked by BT interfaces, the total transmission time to send a packet can be considered as the sum of three times, namely: first, the wired transmission time required to send data from the microcontroller to the BT management unit; second, the latency in the construction of the BT L2CAP packet; and third, the wireless transmission time required to send the BT packet over to the free space. In the proposed case, the first time can be managed by the user, and its rate is selectable from 9600 to 115 200 b/s. The second time depends on the considered BT chip, and it is independent from the selected baud rate; and the third time depends upon the BT standard and the type of connection (ACL and SCO, forward and reverse). For the considered case, the second time is high if it is compared to both the first and third times. This means that it acts as an offset that can from one hand worsen the CPD total value, with respect to the wired case (since it increases the overall synchronization time), and from the other hand, it makes the reduction of the

first time due to the increasing of the transmission baud rate less visible.

Other tests have been executed for the BT wireless transmission channels. In particular, the influence of the BT transmission mode on the TPSN synchronization algorithm has been investigated. Results are reported in Table V. Due to the small size of the synchronization packet, no changes are substantial in the CPD_{MEAN} and σ_{CPD} total values. Very similar results have been observed for the FTSP synchronization protocol. Final tests have been executed for the TPSN algorithm to investigate, in the case of a BT connection, if the role of master or slave of the sender might influence the considered figures of merit. Results have demonstrated that a change in the CPD sign is highlighted when the sender node operates as BT master. These results are not reported in the paper.

Further tests have been executed considering the effect of the Wi-Fi wireless connection. Figs. 7 and 8 report behaviors of the H_{CPD} for both the considered algorithms when the wireless Wi-Fi TCP-IP and UDP connections are considered. Since the adopted wired to Wi-Fi interfaces only allows 115 200 b/s connection, this test has been considered in the setup.

Some considerations can be drawn.

- Once again, the TPSN exhibits a better performance than FTSP for all the considered figures of merit.
- The WiFi UDP connection shows better results than the WiFi TCP-IP one. This can be due to the reduction of transmitted packets required by the UDP protocol.
- The presence of the BT wireless connection worsens the performance respect to WiFi even if they have been constrained at the same wired to wireless baud rate (115 200 kb/s). This is due to a greater latency of the BT connection in respect to WiFi.

Other tests have been executed increasing the complexity of the network (Setup B). Preliminary considerations have to be made to explain the tests carried out. The TPSN has not been tested in this setup, since, whatever the number of nodes, it

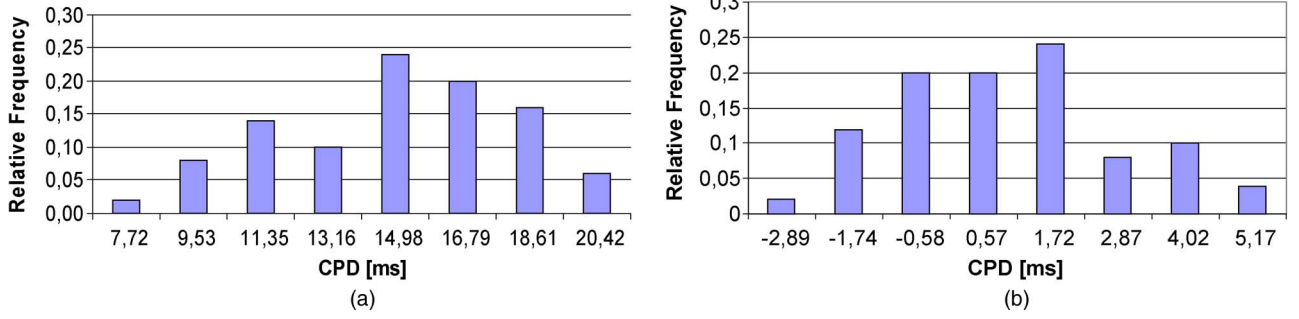


Fig. 8. H_{CPD} histograms for (a) FTSP and the (b) TPSN synchronization algorithms, when the Wi-Fi (UDP) wireless connection is considered.

always realizes a one-to-one synchronization. In addition, it can be highlighted that, in case of N nodes to be synchronized to the sender, the total synchronization time can be considered as the N -times the single TPSN synchronization. Also, the computational burden follows similar consideration. Finally, since the N TPSN synchronizations are executed in different times, because of the different clock drifts and skews, an incremental clock accumulation error may involve all the nodes. As far as the FTSP is concerned, when the number of nodes increases, it shows its better peculiarities since its capability of broadcast synchronization. In this case, if N nodes are present in the network, the FTSP uses the same number of transmissions of a single synchronization. In addition, since the N synchronizations are made in the same time, nonincremental error involves the nodes. For these reasons, only the FTSP performance has been tested. Table VI shows the FTSP performance when five interface nodes are present in the network, and the Wi-Fi (UDP) transmission channel is used.

It is then possible to note that:

- the CPD typically decreases in the receiver to receiver synchronizations. A minimum value of $300 \mu s$, which is closer to what is expected in literature, had been experienced.
- Once again it is demonstrated that the performance of the CPD is strictly related to the accuracy of the clock of the considered nodes. It can be highlighted that since the clocks of Receiver #2, #3 and #4 are very close, the respective CPDs are lower than those experienced to the Receiver #1 node.
- The σ_{CPD} total value is very close to that experienced in Setup A, therefore confirming the general performance of the synchronization protocol.

V. CONCLUSION

Suitable measurement setups and tests for evaluating the performance of synchronization algorithms in small-scale WSNs have been proposed in this paper. The effects of the most persuading factors, such as the clock drift, the latency of radio channels, and the limited bandwidth of the transmission channel have been accounted for.

Tests have been applied to two of the most common synchronization algorithms, the TPSN and the FTSP intentionally developed for a cost-effective WSN, where the computational and memory burdens are important constraints. Results show

TABLE VI
ANALYSIS OF THE FTSP PERFORMANCE WHEN FOUR NODES ARE PRESENT IN THE NETWORK AND THE Wi-Fi (UDP) TRANSMISSION CHANNEL IS USED. THE CPD AND THE σ_{CPD} FIGURES OF MERIT ARE INVOLVED

	Sender	Receiver #1	Receiver #2	Receiver #3	Receiver #4
	[ms]	[ms]	[ms]	[ms]	[ms]
CPD					
Sender		7.40	6.30	6.60	5.80
Receiver #1	7.40		-1.10	-0.80	-1.60
Receiver #2	6.30	-1.10		0.30	-0.50
Receiver #3	6.60	-0.80	0.30		-0.80
Receiver #4	5.80	-1.60	-0.50	-0.80	
σ_{CPD}					
Sender		1.20	1.10	1.50	1.30
Receiver #1	1.20		1.6	1.92	1.77
Receiver #2	1.10	1.6		1.86	1.70
Receiver #3	1.50	1.9	1.86		1.98
Receiver #4	1.30	1.8	1.70	1.98	

that some factors influence the overall performance of the considered protocols. They concern with the synchronization protocols implementation, the Δ_{clock} of the considered nodes, and the latency and bandwidth of the transmission medium.

Further study will have to do with the characterization of more complex protocols, such as the IEEE 1588 PTP, and the estimation of two types of effects, such as the influence of the measurement task on the synchronization task, and the presence of polluted environments, such as those that can be found in industrial applications.

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